Quang-Huy Nguyen

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RESEARCH INTERESTS

My research integrates multi-omics and clinical data using statistical and AI methods for multimodal modeling, uncertainty quantification, agentic reasoning, and federated learning. I use these approaches to uncover disease mechanisms and advance precision risk prediction and stratification across diverse populations.

EDUCATION

Auburn University — **Auburn, AL**
Ph.D. in Computer Science & Software Engineering — *08/2023 – Present*

Auburn University — **Auburn, AL**
M.S. in Computer Science & Software Engineering — *08/2023 – 12/2026*

Dai Nam University — **Hanoi, Vietnam**
B.Sc in Pharmacy — *10/2015 – 08/2020*

RESEARCH EXPERIENCE

Department of Computer Science & Software Engineering, Auburn University — **Auburn, AL**
PhD student — *05/2025 – Present*

Advisors: Professor Wei-Shinn Ku, Research Assistant Professor Zongliang Yue, and Assistant Professor Hao Chen.

- Developed *scBatchProx*, a federated learning-inspired framework for post-hoc single-cell batch integration via latent refinement, enabling continual and cumulative updates under dataset growth.
- Developed *FedWQ-CP*, a federated conformal prediction framework for uncertainty quantification under joint data and model heterogeneity. Achieved robust client-level and global coverage across 7 vision and medical imaging datasets.
- Developed *cORA*, a context-aware pathway-prioritization framework that incorporates experimental conditions and research objectives to identify biologically relevant pathways while reducing generic enrichment signals.
- Developed a Python-based Model Context Protocol (MCP) server integrating the PAGER database with ChatGPT.

Department of Computer Science, Hanoi University of Science and Technology — **Hanoi, Vietnam**
Visiting Graduate Student — *12/2024 – 01/2025*

Advisor: Associate Professor Duc-Hau Le.

- Developed *PAG-Agent*, a web-based, LLM-driven platform integrating bulk and single-cell RNA-seq workflows, including differential expression, pathway enrichment, meta-analysis, literature-assisted interpretation, and hypothesis generation.

Department of Computer Science & Software Engineering, Auburn University — **Auburn, AL**
Graduate Research Assistant — *08/2023 – 08/2024*

Advisors: Research Assistant Professor Phi Hung Bya and Associate Professor Tin Nguyen.

- Conducted methodological reviews of LC-MS–based targeted and untargeted metabolomics and large language models in biomedicine; Dockerized 13 metabolomics software tools as reproducible services to improve accessibility and deployment.
- Led computational analyses of *Candidozyma auris*, including BLAST-based homology mapping and KEGG/GO enrichment to characterize pathways associated with stress adaptation and antifungal resistance; contributed to manuscript preparation, resulting in a co-first-author publication in *Nature Communications*.

School of Computer Science & Engineering, Thuyloi University — **Hanoi, Vietnam**
Visiting Scientist — *10/2021 – 07/2023*

Advisor: Associate Professor Duc-Hau Le.

- Developed *DrGA*, an R toolkit integrating cancer driver-gene analysis, gene-set enrichment, gene–clinical association, module-level analysis, and patient stratification.

School of Computer Science & Engineering, Thuyloi University — **Hanoi, Vietnam**
Undergraduate Scientist — *12/2017 – 07/2019*

Advisor: Associate Professor Duc-Hau Le.

- Replicated published computational studies, designed experiments, and implemented statistical and machine-learning algorithms in R for high-dimensional omics data.

INDUSTRY EXPERIENCE

Center for Biomedical Informatics, Vingroup Big Data Institute — **Hanoi, Vietnam**
Research Scientist — *10/2020 – 09/2021*

Managers: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

- Developed *oCEM*, an R package for detecting overlapping co-expressed gene modules using ICA/IPCA, with a permutation-based procedure for selecting the optimal number of components.
- Designed and implemented a multi-omics analysis pipeline that identified novel prognostic subgroups of breast cancer (log-rank $P = 0.002$ versus $P = 0.598$ for PAM50); led manuscript preparation and writing.
- Co-mentored an undergraduate student on computational clustering and characterization of uveal melanoma patient subgroups.
- Developed *DeepSSC*, a semi-supervised deep-learning framework for multi-omics cancer subtype classification and biomarker discovery; co-mentored 2 undergraduate students on model development.
- Co-designed and executed VN1K's multi-caller consensus SNP/short-indel detection workflow, performed downstream genome-wide association studies and pharmacogenomic analyses. The work was published in *Nature Communications* (2026).
- Led the Vietnamese type 2 diabetes project focused on polygenic risk prediction and mechanism discovery; identified Beta-cell +PI (proinsulin) as the only mechanism significantly associated with T2D in Vietnamese individuals; co-drafted the manuscript.

Research Intern — *03/2020 – 10/2020*

Internship mentor: Associate Professor Duc-Hau Le.

- Developed and implemented an R-based multi-omics pipeline for systematic identification and functional analysis of cancer driver genes.

HONORS AND AWARDS

Next Gen Leadership Award — **Auburn, AL**
AGBT Precision Health — *07/2026*

- Selected as **one of 10 recipients worldwide** from a highly competitive applicant pool spanning academia, healthcare, and industry.
- Invited to deliver a two-minute awardee spotlight and present *PAG-Agent* at the 2026 AGBT Precision Health Meeting.
- Received a **$1,000 full scholarship** covering travel, lodging, and registration.

Benchmarks Resource Grant Program — **Auburn, AL**
Kaggle Research Grant Program — *07/2026*

- Awarded a Benchmarks Resource Grant providing up to **$50/day** and **$500/month** in compute/API credits.

Gold Reviewer Award — **Auburn, AL**
ICML 2026 — *05/2026*

- Recognized as a **Gold Reviewer**, ranking in the **top 25% of eligible reviewers** based on Area Chair ratings.

Google PhD Fellowship in Health Research (University-Level Nomination) — **Auburn, AL**
Auburn University — *04/2026*

- Selected as **one of three Ph.D. students** nominated university-wide by the Auburn University Graduate School for the 2026 Google PhD Fellowship in Health Research.

Certificate of Merit for Academic Excellence and Class Leadership — **Hanoi, Vietnam**
Dai Nam University — *08/2020*

- Ranked in the top 3% of the department's 2015 cohort (nearly 110 students).
- Served as class president for five consecutive years (2015–2020).

27th Mathematics Olympic for Students — **Nha Trang, Vietnam**
Vietnam Mathematical Association — *04/2019*

- Received an **honorable mention** in Linear Algebra among 640 undergraduate competitors representing 80 institutions nationwide.
- Represented my university as its first and only participant and became the first Vietnamese non-engineering student in the history of the competition.

PEER-REVIEWED PUBLICATIONS

†co-first author, *co-corresponding author

11. Trang T. H. Tran, *et al.*, **Quang-Huy Nguyen**, *et al.*, Quan Nguyen*, Van H. Vu*, Nam S. Vo*. *VN1K is a pangenome-informed multi-omics and phenomics resource for the Vietnamese population.* **Nature Communications**, 17:6679. doi:10.1038/s41467-026-75375-0 2026

10. Ching-Lan Chang†, Michael A. Moshi†, **Quang-Huy Nguyen†**, *et al.*, David Hess*, Edwin C. Oh*. *Wastewater intelligence predicts the emergence of clinically-relevant and drug-resistant Candidozyma auris at healthcare facilities.* **Nature Communications**, 17:5415. doi:10.1038/s41467-026-71960-5 2026

9. **Quang-Huy Nguyen**, Ha Nguyen, Edwin C. Oh, Tin Nguyen. *Current approaches and outstanding challenges of functional annotation of metabolites: a comprehensive review.* **Briefings in Bioinformatics**, 25(6):bbae498. doi:10.1093/bib/bbae498 2024

8. Duc-Hau Le, **Quang-Huy Nguyen**, Lan T.M. Dao. *Editorial: Computational and Experimental Approaches in Exploring the Role of Genetics and Genomics in Multifactorial Diseases.* **Frontiers in Genetics**, 13:873069. doi:10.3389/fgene.2022.873069 2022

7. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *DrGA: cancer driver gene analysis in a simpler manner.* **BMC Bioinformatics**, 23:86. doi:10.1186/s12859-022-04606-0 2022

6. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *Identification and validation of a novel three hub long noncoding RNAs with m6A modification signature in low-grade gliomas.* **Frontiers in Molecular Biosciences**, 9:801931. doi:10.3389/fmolb.2022.801931 2022

5. **Quang-Huy Nguyen**, Duc-Hau Le. *oCEM: Automatic detection and analysis of overlapping co-expressed gene modules.* **BMC Genomics**, 23:39. doi:10.1186/s12864-021-08072-5 2022

4. Thi Hai Yen Nguyen†, Tin Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *Re-identification of patient subgroups in Uveal Melanoma.* **Frontiers in Oncology**, 11:731548. doi:10.3389/fonc.2021.731548 2021

3. **Quang-Huy Nguyen**, Duc-Hau Le. *Similarity calculation, enrichment analysis, and ontology visualization of biomedical ontologies using UFO.* **Current Protocols**, 1:e115. doi:10.1002/cpz1.115 2021

2. **Quang-Huy Nguyen**, Duc-Hau Le. *Improving existing analysis pipeline to identify and analyze cancer driver genes using multi-omics data.* **Scientific Reports**, 10:20521. doi:10.1038/s41598-020-77318-1 2020

1. **Quang-Huy Nguyen**, Hung Nguyen, Tin Nguyen, Duc-Hau Le. *Multi-omics analysis detects novel prognostic subgroups of breast cancer.* **Frontiers in Genetics**, 11:574661. doi:10.3389/fgene.2020.574661 2020

PEER-REVIEWED CONFERENCES

1. Hung Nguyen, Bang Tran, Duc Tran, **Quang-Huy Nguyen**, Duc-Hau Le, Tin Nguyen. *Disease subtyping using community detection from consensus networks.* **2020 12th International Conference on Knowledge and Systems Engineering (KSE)**, Can Tho, Vietnam, IEEE, 2020; 318–323. Oral Presentation. doi:10.1109/KSE50997.2020.9287843

BOOK CHAPTERS

1. Zongliang Yue*, Zeru Zhang, **Quang-Huy Nguyen**, Thanh Nguyen, Zhandos Sembay, Hao Chen, Robert Welner, Jake Chen*. *AI-driven gene-sets, networks, pathways, and interactions analyses of multi-omics data.* **Progress in Molecular Biology and Translational Science**, Vol. 221, pp. 277–313, Elsevier. doi:10.1016/bs.pmbts.2026.01.023 2026

WORKS UNDER REVIEW AND PREPARATION

†co-first author, *co-corresponding author

5. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *scBatchProx: Federated-Inspired Refinement for Stable Cell-Type Discriminability under Heterogeneous Batch Compositions.* **arXiv**, 2602.00423. (Under review). doi:10.48550/arXiv.2602.00423 2026

4. **Quang-Huy Nguyen**, Zeru Zhang, Duc-Hau Le*, Jake Y. Chen*, Wei-Shinn Ku*, Hao Chen*, Zongliang Yue*. *PAG-Agent: a biologist-oriented research assistant for context-aware pathway-level analysis and interpretation.* **bioRxiv**, 2026.06.02.729674. (Under review; **Next Gen Leadership Award**; **Kaggle Benchmarks Resource Grant**). doi:10.64898/2026.06.02.729674 2026

3. Hoang Le†, Van-Minh Nguyen†, Duc-Hai Tran, Van-Anh Tran, Van-Nui Nguyen, **Quang-Huy Nguyen**†*, Duc-Hau Le*. *A Semi-Supervised Deep Learning Framework for Multi-Omics Cancer Subtype Classification and Biomarker Discovery.* **bioRxiv**, 2022.01.13.476268. (Under review). doi:10.1101/2022.01.13.476268 2026

2. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity.* **arXiv**, 2602.23296. (Under review). doi:10.48550/arXiv.2602.23296 2026

1. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *FedeKD: Energy-Based Gating for Robust Federated Knowledge Distillation under Heterogeneous Settings.* **arXiv**, 2605.05553. (Under review). doi:10.48550/arXiv.2605.05553 2026

PRESENTATIONS AND TALKS

1. 2026 AGBT Precision Health Meeting (**Next Gen Leadership Award**), The Loews Coronado Bay Resort, San Diego, CA. *Topic: PAG-Agent: A biologist-oriented virtual research assistant for context-aware pathway analysis, prioritization, and interpretation.* 2026

2. 2024 MW CTR-IN Annual Meeting, The Westin Las Vegas Hotel & Spa, Las Vegas, NV. *Topic: A Systems-level Approach for Predicting Patient Survival using Multi-omics Data.* 2024

RESEARCH PROJECTS AND COMPETITIONS

nano-scFMs (Repo) **Auburn, AL**
Independent Open-Source Project *06/2026 – Present*

- Built a collection of lightweight, faithful PyTorch reimplementations of single-cell foundation models to improve readability, reproducibility, and accessibility for AI biology research.
- Reimplemented model architectures, preprocessing pipelines, tokenization workflows, pretrained checkpoint loading, and embedding extraction for single-cell `.h5ad` data.

Cell Maps for AI (CM4AI) CodeFest, UAB **Birmingham, AL**
NIH Bridge2AI Consortium *08/2025*

- Collaborated in a team of 5 to create separate embeddings of yeast fitness data, proteomic mass spectrometry, and cellular imaging, then fuse them into a co-embedding to construct a with-yeast multiscale subcellular hierarchy that uncovered hidden biological processes in triple-negative breast cancer.
- Implemented the analysis pipeline based on the agreed workflow in Python.

The 1000 Vietnamese Genomes Project (VN1K) **Hanoi, Vietnam**
Vingroup Joint Stock Company *2019 – 2024*

PI: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

GenomeAsia100K **Hanoi, Vietnam**
NTU and others *2017 – 2022*

PI: Dr. Stephan Schuster (Singapore side) and Associate Professor Duc-Hau Le (Vietnam side).

TEACHING EXPERIENCE

Department of Computer Science and Software Engineering, Auburn University **Auburn, AL**
Graduate Teaching Assistant for COMP-1220 Python & COMP-1230 Matlab *08/2024 – Present*

- Teaching Python and Matlab fundamentals to ≈ 60 undergraduates and guiding them through problem-solving exercises in lab sections.

PROFESSIONAL SERVICE

- **Conference Reviewer:** ICML 2026, NeurIPS 2026, ICDM 2026, AAAI 2027.
- **Journal Reviewer:** Frontiers in Genetics (2021), Frontiers in Cell and Developmental Biology (2022), Frontiers in Bioinformatics (2026), Frontiers in Artificial Intelligence (2026), Functional & Integrative Genomics (2026), Allergy, Asthma & Clinical Immunology (2026).